LOHITH PAKALA

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Computer Science (AI) undergraduate at IIITDM Kancheepuram with hands-on experience in machine learning, computer vision, and full-stack AI systems. Experienced in building real-world AI applications including surveillance systems, medical imaging models, and LLM-powered tools. Passionate about leveraging AI to solve practical problems and create scalable, impact-driven solutions

EDUCATION

Bachelor of Technology in Computer Science (AI) - 7.69 CGPA **2023-Present**

IIITDM Kancheepuram , Chennai

Higher Secondary - 944/1000 **2021-2023**

Narayana Group Of Education, Vijayawada

LEADERSHIP AND MANAGEMENT EXPERIENCE

Joint Secretary, Student Council IIITDM Kancheepuram **April 2025 - April 2026**

- Coordinated institute-level events, including **Convocation 2025**, ensuring structured stage flow and smooth event execution.
- Organized the **Orientation** Program for incoming students.
- Served as **Vice President of the Techno-Cultural Fest**, leading cross-functional coordination and planning for institute-wide event execution.

TECHNICAL SKILLS

Programming Languages	: Python, C
Databases	: SQL, MySQL
Machine Learning	: Scikit-learn, PyTorch, OpenCV, XGBoost
LLMs & AI	: Whisper, Ollama, LLaMA
Frameworks	: FastAPI, Flask, Streamlit
Tools	: Docker, Git, Linux
Relevant CourseWork	: Pattern Recognition and ML, Deep Learning, Computer Vision, Data Structures, Database Systems

TECHNICAL PROJECTS

1. Bone Age Prediction from Hand Radiographs GitHub

- Built a dual-task deep learning pipeline for bone age estimation on RSNA hand radiographs using PyTorch, combining regression + classification heads; achieved **7.38 MAE**, **R^2 = 0.94**, and **88.9% classification accuracy**.
- Applied Grad-CAM visualizations and XGBoost ensembling to improve interpretability and predictive robustness.

2. Text guided Brain Tumor segmentation using Vision-Language Model GitHub

- Developed a multimodal brain tumor segmentation system integrating FLAIR MRI scans and radiology reports using CLIP, BioClinicalBERT, cross-attention fusion, and FiLM-conditioned U-Net, achieving a Dice score of **0.828** on the BraTS 2020 dataset.
- Designed and evaluated three segmentation architectures (Image-Only U-Net, Concatenation Fusion, Cross-Attention VLM), demonstrating a **13.2% Dice improvement** and **19.5% reduction in HD95** over the image-only baseline through vision-language fusion.

3. Intelligent Surveillance System GitHub

- Designed a robust multi-object tracking pipeline using **YOLOv8** + **ResNet-18 with cosine-similarity-based re-identification** to preserve identity across occlusions.
- Integrated OpenCV-based alerting and monitoring through a PyQt interface for real-time surveillance analysis.

4. AI Customer Feedback Analyzer using Local LLMs GitHub

- Developed a customer feedback analysis application using a local LLM (Mistral via Ollama) for sentiment detection, issue classification, priority assessment, and recommended actions.
- Integrated Flask and SQLite to build an analytics dashboard supporting a fully local, privacy-focused inference pipeline.